CF10: Complete Formalism — VDM Lattice Hydrodynamics, Continuum Limit, and Regularity Program

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Abstract

This Complete Formalism (CF10) specifies a VDM "fluid sector" built from lattice hydrodynamics (lattice Boltzmann / discrete walkers / multi–relaxation-time kernels) expressed in metriplectic form. The document has two tightly separated goals: (i) a discrete-level contract—state space, generators, invariants/entropy, and a gate suite for global stability and hydrodynamic consistency; and (ii) a conditional regularity program for incompressible three-dimensional Navier–Stokes. The regularity component is intentionally framed as a falsifiable hypothesis lattice rather than an analytic proof: the only genuinely hard PDE step is isolated as an explicit A8-style conjecture requiring time-uniform geometric decay of dyadic-shell enstrophy together with scale-wise dominance of dissipation over stretching. Under this conjecture, a short lemma chain (Littlewood–Paley + Bernstein bounds → Beale–Kato–Majda criterion) yields global smoothness. The deliverable is a publishable program specification plus implementation-ready gates (Taylor–Green viscosity recovery, cavity incompressibility, spectrum-tail decay, and refinement collapse) for a companion notebook CFN10.

Keywords lattice Boltzmann method · hydrodynamic limit · Chapman–Enskog expansion · incompressible Navier–Stokes · regularity · enstrophy spectrum · Littlewood–Paley · Beale–Kato–Majda · metriplectic dynamics · GENERIC · Void Dynamics Model

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1 Objective, scope, and non-claims

The VDM program treats numerical dynamics as a measuring instrument: discrete update rules are admitted only if they satisfy explicit invariants, entropy monotonicity, and locality gates aligned with the VDM axiom registry [Lietz, 2025a]. Gap Module **F1** concerns continuum fluids: how a discrete, locally causal lattice dynamics can reproduce incompressible Navier–Stokes in an auditable limit while exposing a falsifiable pathway to three-dimensional regularity.

Two deliverables. CF10 is intentionally split into two cleanly separated deliverables:

1. **Discrete-level contract (engineering deliverable).** A lattice-hydrodynamics sector (lattice Boltzmann / discrete walkers / multi–relaxation-time kernels) is specified in metriplectic form, including state space, generators, and a minimal gate suite for global stability and hydrodynamic consistency.
2. **Conditional regularity program (scientific deliverable).** A single explicit conjecture (Section 5.4) translates the A8 "finite hierarchy" principle into Navier–Stokes language. Under that conjecture, a short lemma chain implies global smoothness via the Beale–Kato–Majda (BKM) criterion.

Scope classification. This module is *derived-limit*: it imports standard results from lattice Boltzmann theory and PDE analysis as intermediate tools, then binds them to VDM observables, gates, and artifact requirements [Lietz, 2025b]. It becomes VDM-axiom relevant only insofar as the companion notebook CFN10 passes the gates specified here.

Non-claims (tight).

- **No Clay-solution claim.** This document does *not* claim a proof of three-dimensional Navier–Stokes global regularity.
- **No novelty claim for classical mathematics.** No novelty is claimed for the lattice Boltzmann method (LBM), the Chapman–Enskog expansion, Littlewood–Paley decompositions, Bernstein inequalities, or the BKM criterion [Succi, 2001, Chen and Doolen, 1998, Bahouri et al., 2011, Beale et al., 1984].
- **Conditional logic only.** Any implication "A8-style spectrum bound $\Rightarrow$ NS regularity" is stated *conditionally* and marked as such.
- **Boundary-condition caution.** The cleanest statements assume periodic domains or well-posed walls; boundary layers are treated as an implementation detail that must be gated (not assumed away).

2 Claim inventory and decisive falsifiers

CF10 treats claims as contracts: each claim is paired with a decisive metric and an explicit falsifier. Claims are numbered **C1–C4**; gates are numbered **F1.1–F1.3**.

2.1 Primary claims

Claim: C1 (Discrete well-posedness and stability of the chosen lattice fluid scheme) *Decisive metric: all stability sub-gates in Gate F1.1 pass under declared parameter ranges.*

Given a declared lattice velocity set (e.g., D2Q9/D3Q27) and collision model (BGK or MRT), the discrete update is globally defined for all time steps under the stated CFL/parameter constraints, preserves nonnegativity (or controlled positivity), and satisfies the relevant invariants/monotonicity checks (mass/momentum conservation; entropy production nonnegativity).

Claim: C2 (Hydrodynamic limit to incompressible Navier–Stokes) *Decisive metric: benchmark residuals and refinement collapse in Gate F1.2.*

In a low-Mach/low-Knudsen regime with declared scaling, the lattice scheme converges to incompressible Navier–Stokes with quantified residuals that shrink under refinement.

Claim: C3 (A8-spectrum gate: practical suppression of blow-up via tail decay) *Decisive metric: time-uniform dyadic-shell tail decay exponent exceeds the analytic critical value in Gate F1.3.*

For VDM-generated three-dimensional flows, the dyadic-shell enstrophy spectrum exhibits geometric decay beyond a dissipation scale and the decay persists uniformly in time under refinement. This is posed as a measurable hypothesis; failure falsifies the regularity program.

Claim: C4 (Conditional analytic implication) *Decisive metric: if Conjecture F1.A holds, Lemmas R1–R3 imply global smoothness.*

If the A8-style conjecture of Section 5.4 holds uniformly in time for incompressible 3D Navier–Stokes, then global regularity follows by the explicit lemma chain in Section 6.

2.2 Assumption ledger

1. **A1 (Metriplectic consistency is checkable).** The chosen discretization admits measurable degeneracy residuals and an entropy production monitor consistent with GENERIC/metriplectic structure [Morrison, 1986, Öttinger and Grmela, 1997, Grmela and Öttinger, 1997, Lietz, 2025b].
2. **A2 (Hydrodynamic topology is declared).** Convergence is asserted only in a declared norm/topology (e.g., grid-converged velocity field; energy decay curve; divergence constraint).
3. **A3 (Spectrum hypothesis is measurable).** The dyadic-shell spectrum and its tail exponent can be estimated reproducibly from resolved fields (FFT + shell binning) with uncertainty quantification.

2.3 Decisive falsifiers

- **Falsifier for C1:** positivity or invariant/monotonicity gates fail under refinement (scheme mis-specified or unstable).
- **Falsifier for C2:** Taylor–Green viscosity recovery error exceeds threshold or does not improve under refinement; incompressibility gate fails.
- **Falsifier for C3:** fitted spectrum-tail exponent falls at or below the analytic critical value (or does not remain bounded away from it uniformly in time).
- **Falsifier for C4:** if C3 fails, the conditional chain cannot be invoked; the program stops (no regularity claim).

3 VDM lattice hydrodynamics: discrete level

3.1 State space and macroscopic moments

Let the spatial domain be a cubic lattice $x \in a\mathbb{Z}^d$ with $d \in \{2, 3\}$ and lattice spacing a. Time is discretized as $t_n = n\,\Delta t$. A finite velocity set $\{c_i\}_{i=0}^{Q-1}$ defines the streaming directions (e.g., D2Q9, D3Q27) [Succi, 2001, Krüger et al., 2017].

The microscopic state is a collection of populations $f_i(x, t_n)$. The standard macroscopic moments are

$$\rho(x, t_n) = \sum_i f_i(x, t_n), \qquad \rho u_\alpha(x, t_n) = \sum_i c_{i,\alpha}\, f_i(x, t_n). \tag{1}$$

Additional moments (stress modes in MRT, or “walker” internal states) may be included provided they satisfy the same gate discipline.

3.2 Update rule as a metriplectic split

A standard LBM update can be written as a streaming step followed by a collision/relaxation step. For clarity, CF10 rewrites the composition as a single-step evolution

$$f_i^{n+1}(x + c_i\Delta t) = f_i^n(x) + \Delta t \Big(J_i[f] + M_i[f] \Big), \tag{2}$$

where

- J is the conservative (streaming / Hamiltonian-like) limb,
- M is the dissipative (collision / metric) limb.

In the single–relaxation-time BGK model, the collision term is

$$M_i[f] = -\frac{1}{\tau}\big(f_i - f_i^{\mathrm{eq}}(\rho, u)\big), \tag{3}$$

with equilibrium populations f_i^{eq} chosen so that the hydrodynamic limit recovers Navier–Stokes [Bhatnagar et al., 1954, He and Luo, 1997, Chen and Doolen, 1998]. MRT variants replace τ with a diagonal relaxation matrix in moment space [d'Humières et al., 2002].

3.3 Generators, invariants, and entropy

The discrete scheme is treated as an instrument whose admissibility is determined by measurable invariants and monotonicity properties.

Conserved moments. For isothermal LBM, the collision operator is designed to conserve mass and momentum:

$$\sum_i M_i[f] = 0, \qquad \sum_i c_i\, M_i[f] = 0. \tag{4}$$

Thermal extensions may additionally conserve total energy.

Entropy (H-functional). An entropy-like functional (minus the classical H-functional) is

$$\Sigma[f] = -\sum_{x,i} f_i(x)\, \ln f_i(x), \tag{5}$$

well-defined when $f_i > 0$. Entropic LBM constructions provide collision maps that guarantee monotone increase of Σ and maintain positivity [Karlin et al., 1999, Ansumali and Karlin, 2002].

Metriplectic constraints. GENERIC/metriplectic structure requires antisymmetry for the Poisson limb and symmetry/positive semidefiniteness for the metric limb [Morrison, 1986, Öttinger and Grmela, 1997, Grmela and Öttinger, 1997]. Operationally, CF10 enforces this via residual gates (degeneracy residuals and entropy production nonnegativity) as defined in the VDM validation metrics registry [Lietz, 2025b].

3.4 Gate F1.1: discrete well-posedness and metriplectic identity checks

Gate: F1.1 (Discrete stability and identity suite) *Threshold: all sub-gates pass with double-precision tolerances (typically* 10^{-12} *scale).*

This gate suite certifies that the discrete update (2) behaves as a stable measuring instrument.

1. **Positivity:** $\min_{x,i} f_i(x, t_n) \ge -10^{-14}$ for all steps (no persistent negative populations).
2. **Mass conservation:** relative drift $\max_n |M_n - M_0| / \max(\varepsilon, |M_0|) \le 10^{-12}$, where $M_n = \sum_x \rho(x, t_n)$.
3. **Momentum conservation:** for periodic/no-slip settings where it applies, $\max_n \|P_n - P_0\| / \max(\varepsilon, \|P_0\|) \le 10^{-12}$.
4. **Entropy production (H-theorem):** per-step $\Delta\Sigma_n \ge -10^{-12}$ for collision-only segments (or the registered entropy-production KPI).
5. **Degeneracy residuals:** $\|J\nabla\Sigma\|_\infty \le 10^{-12}$ and $\|M\nabla I\|_\infty \le 10^{-12}$ for the declared generators I and Σ (registry KPI: degeneracy residuals).

4 Hydrodynamic limit to incompressible Navier–Stokes

4.1 Scaling and Chapman–Enskog expansion

A hydrodynamic limit introduces macroscopic coordinates $x' = ax$ and $t' = n\Delta t$, and assumes a low-Knudsen regime in which gradients are small on the lattice scale. Standard derivations use either diffusive scaling

$\Delta t \sim a^2$ or near-incompressible scalings with small Mach number [Chapman and Cowling, 1970, Succi, 2001, Krüger et al., 2017].

A Chapman–Enskog expansion writes

$$f_i = f_i^{(0)} + \epsilon f_i^{(1)} + \epsilon^2 f_i^{(2)} + \cdots, \tag{6}$$

with $\epsilon \sim \mathrm{Kn}$ (Knudsen number), and enforces solvability conditions at each order. With a standard polynomial equilibrium and BGK/MRT collision, the leading orders yield continuity and momentum equations with viscosity

$$\nu = c_s^2\Big(\tau - \tfrac{1}{2}\Big)\Delta t, \tag{7}$$

where c_s is the lattice sound speed [Succi, 2001, He and Luo, 1997, Krüger et al., 2017].

In the incompressible limit ($\rho \approx \rho_0$, $\nabla \cdot u = 0$), the target PDE is

$$\partial_t u + (u \cdot \nabla)u = -\frac{1}{\rho_0}\nabla p + \nu\,\Delta u. \tag{8}$$

4.2 Gate F1.2: hydrodynamic consistency and benchmark recovery

Gate: F1.2 (LBM→NS benchmark gates) *Threshold: Taylor–Green viscosity recovery rel. error* ≤ 0.05*; lid-cavity divergence max* $\leq 10^{-6}$.

This gate operationalizes "LBM recovers Navier–Stokes" using canonical fluid benchmarks and the VDM validation metric registry [Lietz, 2025b].

1. **Taylor–Green viscosity recovery.** Use the Taylor–Green vortex decay to estimate ν from the energy decay curve and require $\mathrm{rel\_err}_\nu \leq 0.05$ at baseline resolution (typically $\geq 256^2$ in 2D). This KPI is registered as "Taylor-Green Viscosity Recovery Error".
2. **Lid-driven cavity incompressibility.** In a lid-cavity benchmark, require $\max_t \|\nabla \cdot u\|_2 \leq 10^{-6}$ (double precision target). This KPI is registered as "Lid Cavity Divergence Maximum" and may be cross-checked against classical cavity references [Ghia et al., 1982].
3. **Refinement collapse.** Under doubled resolution (and correspondingly adjusted Δt when required), the benchmark residuals must decrease in a manner consistent with the declared scheme order (reported as a log–log slope and R^2).

5 Hierarchical cascade and an A8-style bound

The open analytic step is to prevent enstrophy from cascading to arbitrarily small scales in three dimensions. CF10 records an A8-style statement that is both (i) strong enough to imply regularity, and (ii) directly measurable in simulation.

5.1 Dyadic shell decomposition and enstrophy

Let $\omega = \nabla \times u$ be vorticity. On a periodic domain, a Littlewood–Paley decomposition writes

$$u = \sum_{k\in\mathbb{Z}} u_k, \qquad u_k = \mathcal{P}_k u, \tag{9}$$

where $\mathcal{P}_k$ projects onto wavenumbers $|\xi| \sim 2^k$ [Bahouri et al., 2011]. Define shell vorticity $\omega_k = \nabla \times u_k$ and shell enstrophy

$$\Omega_k(t) = \int |\omega_k(x,t)|^2\,dx = \|\omega_k(\cdot,t)\|_2^2. \tag{10}$$

5.2 Dissipation cost at scale k

The Navier–Stokes energy balance has dissipation rate

$$\frac{dE}{dt} = -\nu\,\|\nabla u\|_2^2. \tag{11}$$

Because $\|\nabla u_k\|_2^2 \sim 2^{2k}\,\|u_k\|_2^2$ on a dyadic shell, and $\|\nabla u_k\|_2$ is comparable to $\|\omega_k\|_2$, one expects a scale-wise dissipation weight

$$\epsilon_k(t) \;\sim\; \nu\, 2^{2k}\, \Omega_k(t). \tag{12}$$

This expresses the "interface-area cost" heuristic: pushing activity to higher k pays a quadratic derivative tax.

5.3 Stretching versus dissipation

The vorticity equation is

$$\partial_t \omega = \nabla \times (u \times \omega) + \nu\, \Delta\omega, \tag{13}$$

where the nonlinear term can stretch vorticity and transfer enstrophy to higher wavenumbers. A scale-local balance heuristic writes

$$\frac{d\Omega_k}{dt} = \underbrace{\mathcal{T}_k[u]}_{\text{stretching/transfer}} - \nu\, 2^{2k}\, \Omega_k \;+\; (\text{commutator terms}). \tag{14}$$

CF10 does not claim a proof of a sharp bound on $\mathcal{T}_k$; instead, it elevates the required inequality to an explicit conjecture.

5.4 Conjecture F1.A

Claim: Conjecture F1.A (Hierarchical cascade bound) *Decisive metric: time-uniform tail decay and per-shell dominance of dissipation over transfer.*

There exist constants $k_0 \in \mathbb{Z}$ and $\beta > 3$, and finite constants C, c_2, such that for any smooth incompressible 3D Navier–Stokes solution with finite initial energy,

1. **Time-uniform shell decay:** for all $t \geq 0$ and all $k \geq k_0$,

$$\Omega_k(t) \leq C\, 2^{-\beta k}. \tag{15}$$

2. **Stretching bounded by dissipation (scale-wise):** the transfer contribution to $d\Omega_k/dt$ satisfies

$$\mathcal{T}_k[u](t) \leq c_2\, \nu\, 2^{2k}\, \Omega_k(t) \qquad (k \geq k_0). \tag{16}$$

Interpretation as A8. This conjecture is the Navier–Stokes-side translation of the A8 "finite-depth hierarchy" heuristic formalized in CF03 and the associated robustness note [Lietz, 2025c,d]. Each cascade step to higher k increases gradient "interface" content, and viscosity taxes that content quadratically. A time-uniform tail bound forbids an "infinite cascade" that would concentrate vorticity at ever smaller scales.

5.5 Gate F1.3: spectrum-tail and time-uniformity checks

Gate: F1.3 (A8-spectrum gate) *Threshold: $\beta_{\min} \geq 3.2$ with tail-fit $R^2 \geq 0.98$, uniformly in time and stable under refinement.*

The companion notebook CFN10 must estimate $\Omega_k(t)$ on resolved fields, fit a tail exponent $\beta(t)$ on a declared tail window $k \in [k_{\text{tail}}, k_{\max}]$, and report $\beta_{\min} = \inf_{t\in\mathcal{T}} \beta(t)$ over a declared time window $\mathcal{T}$. The gate requires:

1. **Tail fit quality:** $R^2 \geq 0.98$ for the log–linear fit $\log_2 \Omega_k \approx a - \beta k$.
2. **Criticality margin:** $\beta_{\min} \geq 3.2$ (a ≈ 0.2 safety margin over the analytic threshold $\beta > 3$).
3. **Refinement stability:** $\beta_{\min}$ does not decrease under refinement (higher $k_{\max}$) beyond statistical uncertainty.

Failure of this gate falsifies C3 and blocks the conditional regularity claim.

6 Conditional regularity theorem and lemma chain

Theorem 1 (Conditional NS regularity via Conjecture F1.A)**.** *Assume Conjecture F1.A holds for an incompressible 3D Navier–Stokes solution with viscosity $\nu > 0$ and finite-energy initial data. Then:*

1. *The enstrophy $\Omega(t) = \|\omega(\cdot, t)\|_2^2$ remains finite for all $t \geq 0$.*
2. *The vorticity supremum norm $\|\omega(\cdot, t)\|_\infty$ is integrable on any finite time interval.*
3. *By the BKM criterion, the solution remains smooth for all time (no finite-time singularity).*

Lemma chain (explicit). Only Conjecture F1.A is nontrivial. The remaining steps are standard and itemized for auditability.

Lemma 1 (R1: tail decay implies bounded enstrophy)**.** *If $\Omega_k(t) \leq C\,2^{-\beta k}$ for all $k \geq k_0$ with $\beta > 1$, uniformly in time, then $\Omega(t) = \sum_k \Omega_k(t)$ is finite and uniformly bounded.*

Sketch. Split $\sum_k$ into a finite low-frequency part plus the tail $\sum_{k \geq k_0} C2^{-\beta k}$, which converges when $\beta > 1$. ☐

Lemma 2 (R2: tail decay implies $\|\omega\|_\infty$ bound via Bernstein)**.** *If $\Omega_k(t) \leq C\,2^{-\beta k}$ for all $k \geq k_0$ with $\beta > 3$, uniformly in time, then $\|\omega(\cdot, t)\|_\infty$ is uniformly bounded.*

Sketch. By Bernstein on a dyadic shell in 3D, $\|\omega_k\|_\infty \lesssim 2^{3k/2}\|\omega_k\|_2 = 2^{3k/2}\Omega_k^{1/2}$. Summing over $k \geq k_0$ gives $\|\omega\|_\infty \leq \sum_k \|\omega_k\|_\infty \lesssim \sum_{k \geq k_0} 2^{3k/2}\,2^{-\beta k/2}$, which converges when $\beta > 3$ [Bahouri et al., 2011]. ☐

Lemma 3 (R3: Beale–Kato–Majda criterion)**.** *If $\int_0^T \|\omega(\cdot, t)\|_\infty\,dt < \infty$, then no finite-time blow-up occurs on $[0, T]$.*

Citation. This is the classical BKM criterion (originally stated for Euler; corresponding statements for Navier–Stokes follow by standard adaptation) [Beale et al., 1984, Constantin and Foiaş, 1988]. ☐

Remark 1 (R4: discrete-to-continuum consistency)*.* Given Gate F1.1 (instrument stability) and Gate F1.2 (hydrodynamic benchmark recovery), the lattice scheme is treated as a convergent approximation to (8) in the declared regime. Uniform bounds implied by R1–R3 are intended to support upgrading a weak limit to a smooth limit when applicable. This step is methodological rather than claimed as a new theorem.

7 Companion notebook CFN10 deliverables (minimum)

The companion notebook is required to emit machine-readable artifacts and pass/fail JSON for each gate.

- **Gate F1.1 artifacts:** positivity/min-population time series; mass and momentum drift logs; entropy production time series; degeneracy residual report (JSON + CSV).
- **Gate F1.2 artifacts:** Taylor–Green energy decay curve with fitted ν and relative error; lid-cavity divergence time series; refinement plots with slopes and R^2.
- **Gate F1.3 artifacts:** dyadic-shell enstrophy spectra $\Omega_k(t)$ sampled over time; tail-fit coefficients (a, β, R^2) per snapshot; $\beta_{\min}$ report with uncertainty and refinement comparison.

8 Integration map and limitations

8.1 Integration map

- **A8 hierarchy inputs:** CF03 provides the general hierarchical partition program and motivates the “finite-depth cascade” lens used in Conjecture F1.A [Lietz, 2025c,d].
- **Fluid validation metrics:** Taylor–Green and lid-cavity KPIs are registered in the VDM validation metrics canon and used directly as Gate F1.2 thresholds [Lietz, 2025b].
- **Downstream use:** once Gates F1.1–F1.2 pass, the fluid sector becomes an admissible VDM instrument for transport, mixing, and nonequilibrium modules that require trustworthy continuum limits.

8.2 Limitations

- **Conjecture dependence:** the only path to a 3D regularity claim runs through Conjecture F1.A. If Gate F1.3 fails, the program halts without interpretive escape hatches.
- **Finite resolution:** any spectrum-tail gate must be stress-tested under refinement to avoid mistaking numerical dissipation for physical tail decay.
- **Boundary layers:** wall-bounded flows require careful gating of boundary-condition implementations; CF10 focuses on periodic/standard cavity baselines.

9 Provenance

This build is intended for Zenodo-style archival.

- Canon snapshot commit (from provenance manifest): `91332f4cf89a52d239eaab6659b8df8d8b9f3202`.
- Source basis: `Derivation/Complete-Formalisms/CF10_Lattice_Fluids_Continuum_and_Regularity_Program.md` (repository snapshot; LaTeX adapted for publication).
- Canon registries referenced: axioms and validation metrics [Lietz, 2025a,b].

End of CF10.

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